Towards Improving Semantic Parsing using Statistical Word Sense Disambiguation

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Outline

GOAL: Provide statistical word sense priors to the TRIPS parser to improve parser accuracy.
DIRECTION:

- Word Sense Disambiguation (**WSD**) errors in the TRIPS semantic parsing can cascade errors
- Providing better ranking on hypotheses may help parser achieve better **WSD** results
- We examine the impact of hints from a WordNet[1]-based **WSD** system on the parser.

The TRIPS Semantic Parser

- Handcrafted grammar and ontology with semantic roles and restrictions
- Sentence preprocessing provides candidate senses to the parser for each word
- Bottom-up chart parser finds best spanning parse given constituent hypotheses
- Heuristic pruning keeps search space manageable

Word Senses in TRIPS

- Single-inheritance semantic hierarchy with features and semantic templates
- [Semantic types + lexical entries] for combined semantic-syntactic information for spans

Rich Mappings in WordNet

- A hand-built lexical database containing 155k words organized into 117k synonym sets (synsets)
- Synsets are related into a hierarchy through hypernym/hyponym relations (amongst others)

TRIPS-WordNet

When using TRIPS and WordNet as a single joined resource, we define the ancestor relation of a synset as follows: if a synset is mapped to a TRIPS type, `anc` returns that type. Otherwise, `anc` returns the synsets WordNet hypernyms. For a given lemma l:

- $\texttt{Int}_{wn}(l)$: the set of WordNet synsets that could represent l
- $\texttt{Int}_{trips}(l)$: the set of TRIPS types with explicit mappings to l

Decoding

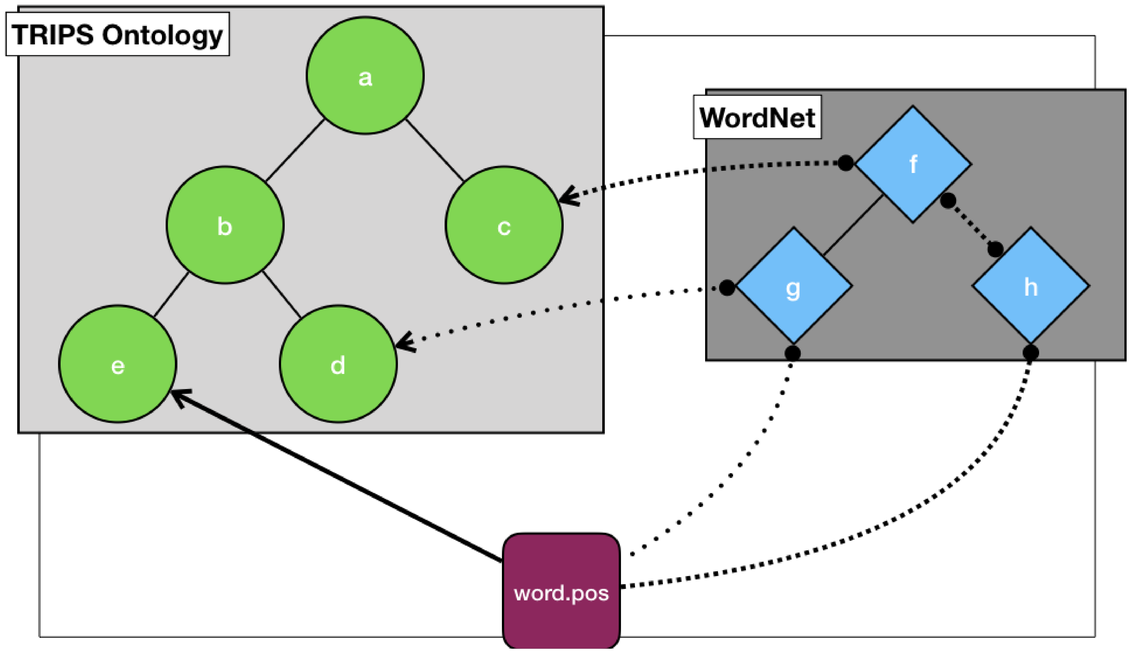
In order to evaluate the changes to the parser, we need to retrieve the WordNet sense from TRIPS types. We select a synset *syn* from $\texttt{Int}_{wn}(l)$.
PTR: $\texttt{argmin}\,\texttt{dist}(t, \texttt{syn}) | t \in ancestors(syn)$, where t is the TRIPS type attached to the lemma l.
WuPalmer: $\texttt{argmin}\,\texttt{WUP}(t, \texttt{syn})$,

WuPalmer Similarity[4]

$$wup(a,b) = \frac{2 * depth(LCS(a,b))}{depth(a) + depth(b)}$$

Evaluation

- Unified WSD test framework [3]
- Compare to **SUPWSD** [2] and Most Frequent Sense (**MFS**) baselines
- Parser is seeded with hints from the top senses in **SUPWSD** output in addition normal constituents
- Hints are added to the chart, ensuring the correct sense is considered in the parsing process
- Resulting parses are decoded into SemEval format for testing in WSD test framework



When using TRIPS and WordNet as a joined resource, we define the *ancestor relation* of a synset as follows: if a synset mapps to a TRIPS type, `anc` returns that type; else, `anc` returns the synset's WordNet hypernyms. Many lemmas have direct mappings to the TRIPS ontology. Additionally, we look at all the possible WordNet synsets for the lemma. We recurse up the combined hierarchy until a TRIPS type is found.

Preliminary Results

SemEval2015

supwsd	MFS	plain	hinted	
711	693	236	217	**ptr**
		458	414	**wup**

Figure 1: the number of correct decisions made by each strategy. [In total, **SEMEVAL2015** has 1022 instances.]

PTR possible

supwsd	MFS	plain	hinted	
577	559	236	217	**ptr**
		381	348	**wup**

Figure 2: the number of correct answers from each strategy, using only instances where the gold sense is reachable via **PTR**.

Observations and Problems

- Mapping a TRIPS type and lemma pair back to a WordNet sense is not straightforward
  - WordNet synsets are not always organized according to the same principles as TRIPS types
  - WordNet is organized by semantics, while TRIPS is a mix of syntax and semantics
  - The **WUP** strategy performs significantly better than **PTR** even when the latter would be theoretically able to find the correct answer
- Alignments between the original sentence and final parse are not always apparent
  - Multi-word-expressions are split up in different ways between TRIPS and WordNet
  - there is not always a one-to-one mapping between nodes in the final parse and words in the sentence
- Current strategy may add confusion
  - Reduction in accuracy through hinting indicates the current hinting strategy may simply add more noise
  - Applying hints during pruning could provide a more dynamic and robust strategy
- Low variability with hinting shows strong preference for semantic restrictions

Next Steps

- Manual evaluation of hinted parses
- A measure of semantic similarity when evaluating results may yield better insights into results
- Alternative word-sense hinting methods, including dynamically guiding pruning strategies on the chart
- More in-depth decoding methods

References

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